Cloud Infrastructure for a Startup Company

Multi-VPC architecture on AWS with a central Transit Gateway

Region: ap-south-1 (Mumbai) | Prepared by Ziad Nady | 2 October 2026 | NTI project

1. Project overview

This project builds a cloud infrastructure on AWS for a startup company. It supports three teams (Admin, ERP and App) in isolated networks that are connected through a single AWS Transit Gateway. The infrastructure hosts a multi-tier ERP system with monitoring and disaster recovery, and a highly available internal application whose servers share documents. The design is cost-conscious: expensive components were avoided or sized down where the trade-off was acceptable for a startup (see section 10).

Item	Details
Purpose	Internal ERP platform and internal application for a startup company
Teams supported	Admin, ERP and App
AWS Region	Asia Pacific (Mumbai), ap-south-1
Network model	Three isolated VPCs interconnected through a Transit Gateway
Availability	Subnets across multiple Availability Zones
Compute	Amazon EC2 with Auto Scaling behind an Application Load Balancer
Shared storage	Amazon EFS (encrypted, automatic backups, lifecycle policy)
Monitoring	Amazon CloudWatch dashboard and alarms, Amazon SNS e-mail notifications
Disaster recovery	RDS backups and snapshots, Lambda automation, Golden AMI

2. Requirements and solutions

#	Requirement	Solution
1	Core infrastructure supporting three teams, each with its own permissions	Three isolated VPCs, Bastion Host, Transit Gateway, IAM groups and policies
2	ERP hosted on AWS with multi-tier design, monitoring and disaster recovery	EC2 web tier (public) and RDS MySQL (private); CloudWatch and SNS; RDS snapshots; Lambda backup; Golden AMI
3	Internal application with high availability and shared documents on all servers	Application Load Balancer, Auto Scaling Group, Amazon EFS
4	Secure administrative access	Bastion Host in the Admin VPC, IAM groups, MFA for privileged users, Security Groups

3. Architecture

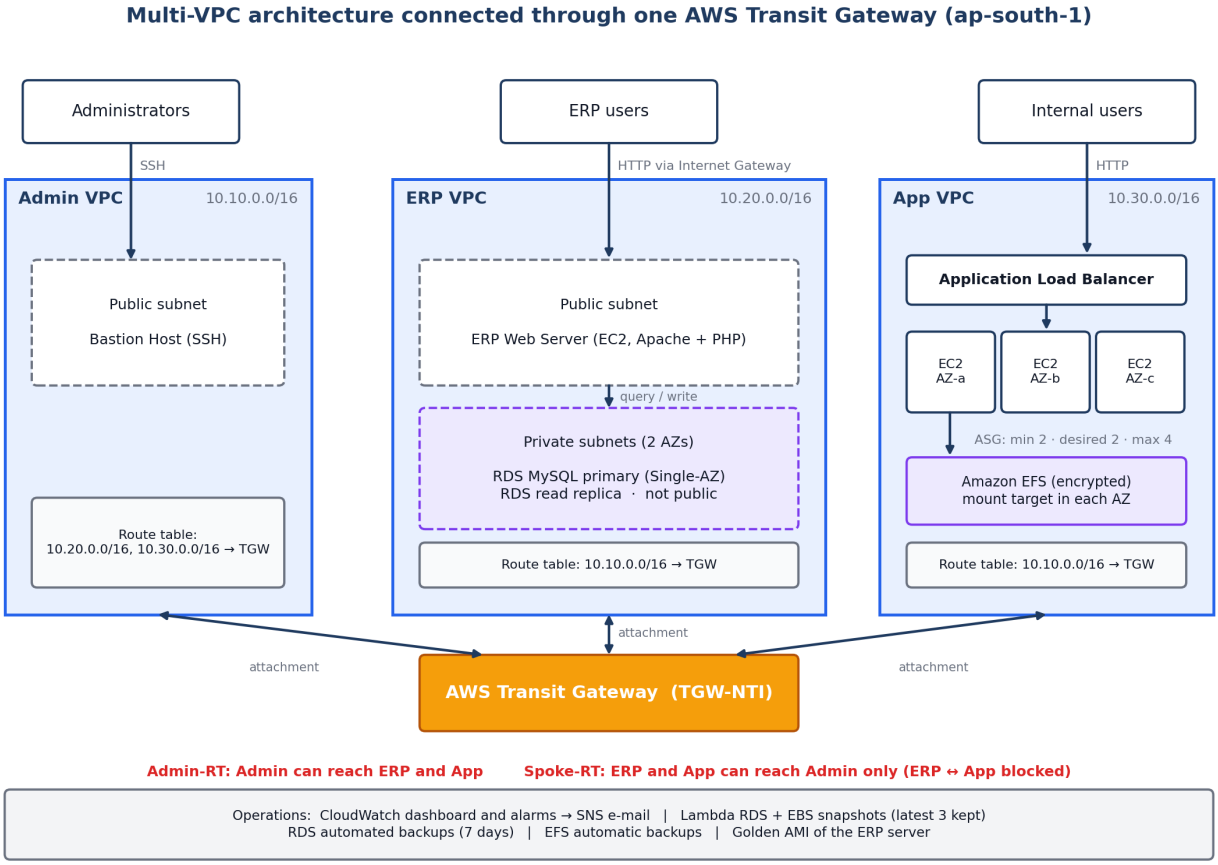


Figure 1: Architecture. Three VPCs (Admin, ERP, App) attach to one Transit Gateway. The App VPC holds the ALB, Auto Scaling Group and EFS; the ERP VPC holds the web server (public subnet) and RDS MySQL (private subnets); the Admin VPC holds the Bastion Host.

Traffic flow. Internal users reach the Application Load Balancer, which forwards to the Auto Scaling Group (EC2); the servers read and write shared documents on EFS. ERP users reach the ERP web server, which queries RDS MySQL. Administrators connect to the Bastion Host in the Admin VPC and reach ERP and App resources through the Transit Gateway.

3.1 Network ranges and main services

Network	CIDR	Contents
Admin VPC	10.10.0.0/16	Bastion Host for administration
ERP VPC	10.20.0.0/16	ERP web server (public subnet), RDS database (private subnets)
App VPC	10.30.0.0/16	ALB, Auto Scaling Group, EFS

Service	Purpose
Amazon VPC	Network isolation, subnets, routing and security boundaries
AWS Transit Gateway	Centralized connectivity between the VPCs
Internet Gateway	Internet connectivity for public resources
S3 Gateway Endpoint	Free private access to Amazon S3 from inside the VPC
Application Load Balancer	Distributes application traffic across EC2 instances

Service	Purpose
Amazon EC2 and Auto Scaling	ERP web server and application servers with automatic capacity adjustment
Amazon RDS (MySQL)	Database tier of the ERP system: one primary instance and one read replica
Amazon EFS	Shared documents accessible by multiple servers
Amazon CloudWatch	Metrics, dashboard, logs and alarms
Amazon SNS	E-mail notifications for alarms and scaling events
AWS Lambda	Automated RDS and EBS snapshots
AWS IAM	Team-specific access and permissions
Bastion Host	Controlled administrative SSH access

4. Network design: Transit Gateway

A single Transit Gateway acts as the central communication point. Each VPC connects through its own attachment, which avoids a mesh of peering connections and makes it easy to add new environments later.

4.1 Transit Gateway configuration

Setting	Value
Name	TGW-NTI (Admin, ERP, App)
Attachments	One VPC attachment per VPC (Admin, ERP, App)
DNS support	Enabled
Default route table association / propagation	Disabled; custom route tables are used for isolation (see 4.2)
Auto accept shared attachments	Disabled

4.2 Keeping ERP and App isolated

A Transit Gateway allows full connectivity between attachments by default, so isolation must be designed explicitly. Two Transit Gateway route tables are used:

TGW route table	Associated with	Propagated routes from	Result
Admin-RT	Admin VPC attachment	ERP and App attachments	Admin can reach both ERP and App
Spoke-RT	ERP and App attachments	Admin attachment only	ERP and App can reach Admin, but not each other

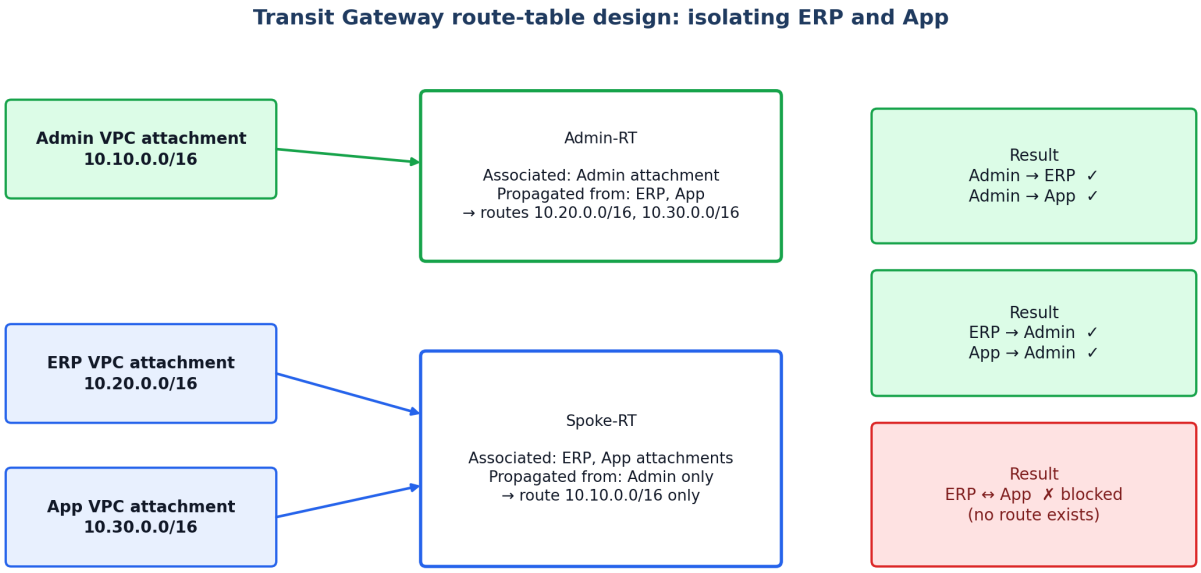


Figure 2: Transit Gateway route-table design. ERP and App share Spoke-RT, which only contains the Admin route, so no path exists between them.

Security Groups add a second layer of control on top of the routing.

4.3 VPC routing

The VPCs use non-overlapping CIDR ranges, each with an Internet Gateway, route tables and subnets across multiple Availability Zones. Each VPC route table sends traffic for the other permitted ranges to the Transit Gateway: the Admin VPC routes 10.20.0.0/16 and 10.30.0.0/16, while the ERP and App VPCs route only 10.10.0.0/16 (Admin). Separate public and private route tables are used, so only public subnets route directly to the Internet Gateway.

4.4 Cost note

A Transit Gateway costs more than VPC Peering: it is billed per attachment-hour plus a per-GB data processing charge (see the estimate in section 10). It was chosen deliberately for central management, scalability and explicit routing control, and it is the largest fixed cost in this design. For a permanently fixed set of three VPCs with very low traffic, VPC Peering would be cheaper.

5. Security and access control

- **IAM groups:** separate groups for the Admin, ERP and App teams. Admin has full access; the other teams have limited access limited to what they need.
- **IAM role for servers:** SSM and CloudWatch Agent permissions.
- **MFA:** multi-factor authentication is enabled for privileged IAM identities.
- **Private resources:** the database is not exposed to the Internet (public access disabled).
- **Security Groups:** ERP-SG allows HTTP 80 and SSH 22, with SSH limited to the Admin VPC range (10.10.0.0/16) because security-group references cannot be used across VPCs connected through a Transit Gateway. DB-SG allows MySQL 3306 from ERP-SG only. The App security groups allow HTTP from the ALB only.
- **Bastion Host:** administrative SSH access goes through the Bastion Host in the Admin VPC and reaches the other VPCs through the Transit Gateway.
- **Data protection in the application:** passwords hashed with password_hash (BCRYPT), prepared statements against SQL injection, and PDO in exception mode.

6. Implementation

6.1 ERP system (multi-tier)

- **Web tier:** Amazon Linux 2023 EC2 with Apache and PHP (php-mysqlnd, php-pdo) in a public subnet.
- **Data tier:** RDS MySQL primary (db.t3.micro, Single-AZ) in private subnets with public access disabled.
- **Read replica:** `erp-mysql-db-replica` is an RDS read replica of the primary. It keeps a second, continuously updated copy of the data, can serve read-only or reporting queries without loading the primary, and can be promoted to a standalone instance as an additional recovery option. It uses the same private subnets and security group rules (not public).
- **Application:** a login page and a welcome page that read and write RDS (tables `users`, `login_log`, `notes`).
- **Domain:** DuckDNS with a cron job (`duck.sh`) updating the public IP every 5 minutes.
- **Transport:** the lab environment serves HTTP; HTTPS with ACM is the first production improvement (section 13).



Figure 3: ERP web application: welcome page after a successful login.

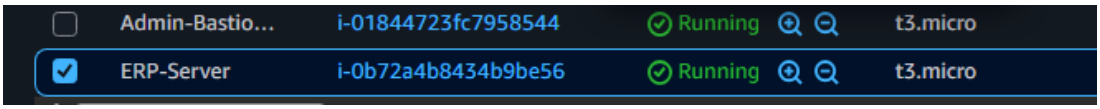


Figure 4: EC2 instances: the Admin Bastion Host and the ERP server, both running (t3.micro).

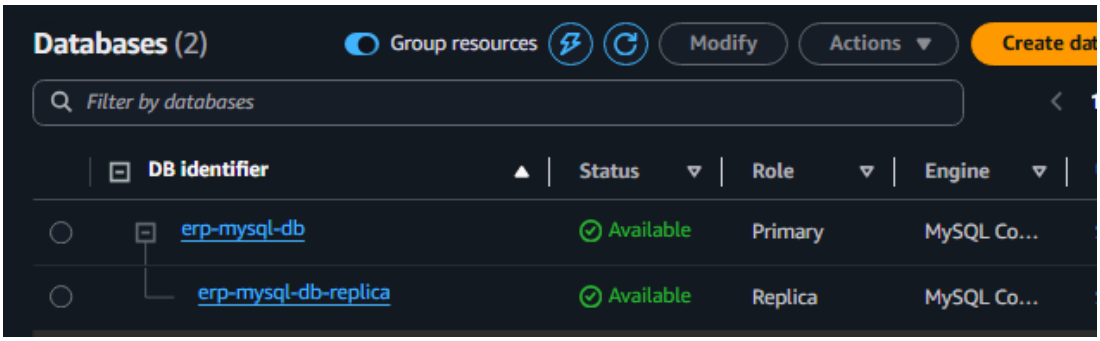


Figure 5: RDS databases: erp-mysql-db (Primary) and erp-mysql-db-replica (Replica), both Available.

6.2 Internal application (high availability)

- **ALB:** distributes requests across healthy servers and performs health checks. Unhealthy instances are removed from rotation automatically.

- **Launch Template:** every new instance is created from the same template (AMI, instance type, security group, user data), so all servers have an identical configuration.
- **Auto Scaling Group:** desired 2, minimum 2, maximum 4 across 3 AZs. Target tracking on CPU at 60%, plus a RAM alarm using the CloudWatch Agent.
- **EFS:** mounted at `/mnt/efs` on every server through the Launch Template user data, so a file written on one server appears on all the others.

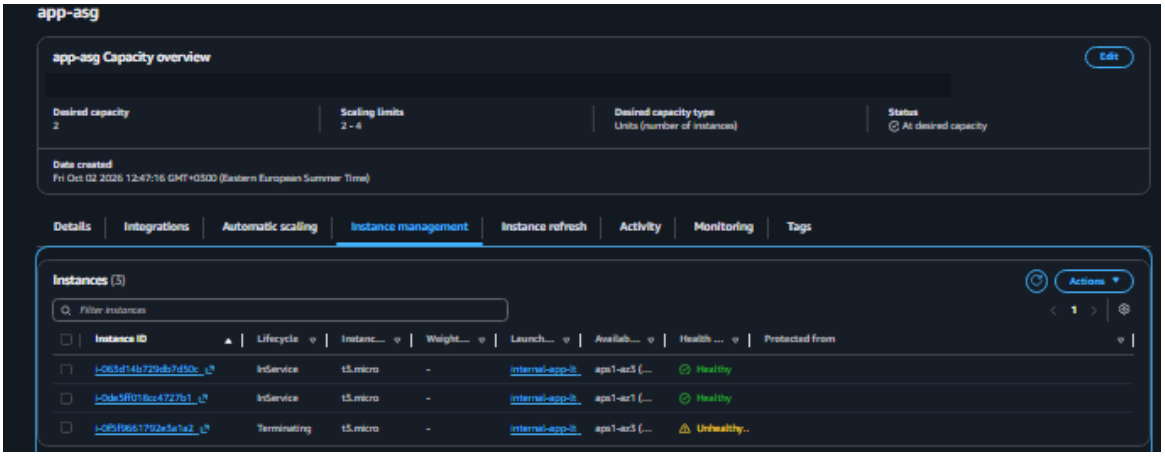


Figure 6: Auto Scaling Group app-asg (desired capacity 2, limits 2 to 4). Two instances are InService and Healthy while one instance is Terminating.

6.3 Shared storage (Amazon EFS)

Setting	Value
Performance mode	General Purpose
Throughput mode	Elastic
Availability	Regional (multi-AZ), mount target in each AZ
Encryption	Enabled (AWS-managed key)
Automatic backups	Enabled
Lifecycle management	Infrequent Access after 30 days; Archive after 90 days

7. Monitoring and alerting

7.1 CloudWatch dashboard

A dedicated CloudWatch dashboard shows in one place: CPU utilization of the ERP server and the Auto Scaling Group, memory usage (from the CloudWatch Agent), ALB request count and 5XX errors, healthy host count, and RDS CPU and free storage.

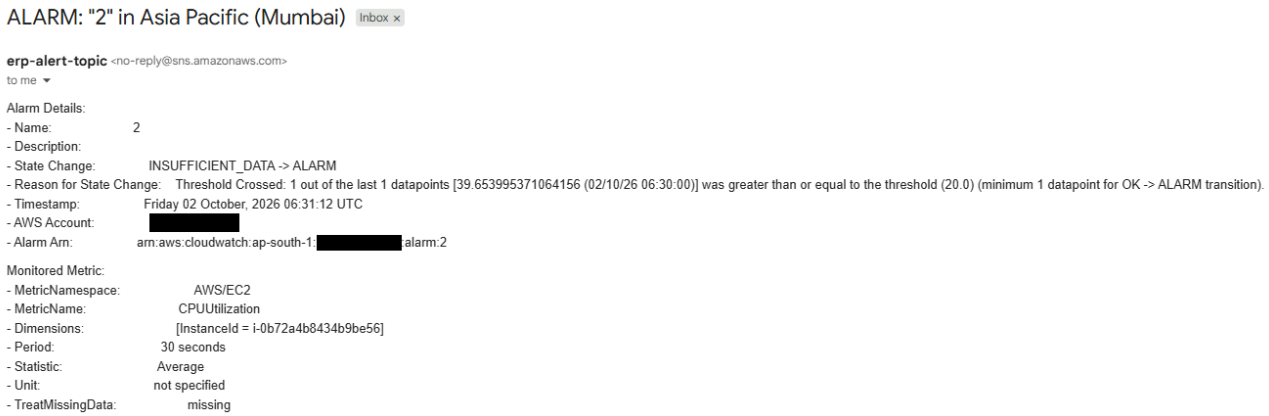
7.2 CloudWatch alarms

Item	Tool	Condition
Server CPU	ASG target tracking (60%) and a CloudWatch alarm on the ERP server	Above threshold
Memory	CloudWatch Agent (`mem_used_percent`)	Above 75% scale out; below 30% scale in
Application errors	Metric filter on `/var/log/httpd/error_log`	Pattern ERROR appears
ALB and RDS	CloudWatch alarms	5XX count, unhealthy hosts, CPU, free storage

The Auto Scaling policy creates two target-tracking alarms (AlarmHigh for scale-out and AlarmLow for scale-in). AlarmLow can show *In alarm* under light load; this is expected and means the group is at its minimum size.

7.3 Notifications with Amazon SNS

An SNS standard topic with a confirmed e-mail subscription receives all alarms and scaling events and e-mails the team.

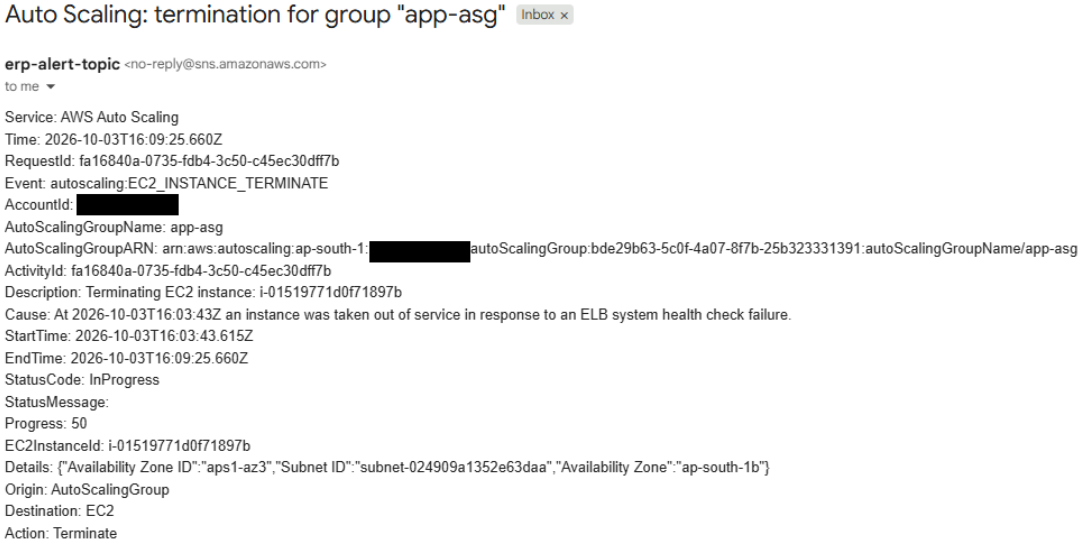

ALARM: "2" in Asia Pacific (Mumbai) Inbox ×

erp-alert-topic <no-reply@sns.amazonaws.com>
to me

Alarm Details:
- Name: 2
- Description:
- State Change: INSUFFICIENT_DATA -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [39.653995371064156 (02/10/26 06:30:00)] was greater than or equal to the threshold (20.0) (minimum 1 datapoint for OK -> ALARM transition).
- Timestamp: Friday 02 October, 2026 06:31:12 UTC
- AWS Account: [redacted]
- Alarm Arn: arn:aws:cloudwatch:ap-south-1:[redacted]:alarm:2

Monitored Metric:
- MetricNamespace: AWS/EC2
- MetricName: CPUUtilization
- Dimensions: [InstanceId = i-0b72a4b8434b9be56]
- Period: 30 seconds
- Statistic: Average
- Unit: not specified
- TreatMissingData: missing

Figure 7: Amazon SNS e-mail for a CloudWatch CPU alarm on the ERP server (state change INSUFFICIENT_DATA to ALARM).


Auto Scaling: termination for group "app-asg" Inbox ×

erp-alert-topic <no-reply@sns.amazonaws.com>
to me

Service: AWS Auto Scaling
Time: 2026-10-03T16:09:25.660Z
RequestId: fa16840a-0735-fdb4-3c50-c45ec30dff7b
Event: autoscaling:EC2_INSTANCE_TERMINATE
AccountId: [redacted]
AutoScalingGroupName: app-asg
AutoScalingGroupARN: arn:aws:autoscaling:ap-south-1:[redacted]autoScalingGroup:bde29b63-5c0f-4a07-8f7b-25b323331391:autoScalingGroupName/app-asg
ActivityId: fa16840a-0735-fdb4-3c50-c45ec30dff7b
Description: Terminating EC2 instance: i-01519771d0f71897b
Cause: At 2026-10-03T16:03:43Z an instance was taken out of service in response to an ELB system health check failure.
StartTime: 2026-10-03T16:03:43.615Z
EndTime: 2026-10-03T16:09:25.660Z
StatusCode: InProgress
StatusMessage:
Progress: 50
EC2InstanceId: i-01519771d0f71897b
Details: {"Availability Zone ID":"aps1-az3","Subnet ID":"subnet-024909a1352e63daa","Availability Zone":"ap-south-1b"}
Origin: AutoScalingGroup
Destination: EC2
Action: Terminate

Figure 8: Amazon SNS e-mail for an Auto Scaling termination event: an app-asg instance was taken out of service after an ELB health check failure.

8. Disaster recovery

- **RDS automated backups:** 7-day retention with point-in-time restore.
- **Manual snapshots:** a Python Lambda function creates RDS snapshots and EBS snapshots (for instances tagged `Backup=true`), keeps only the latest 3, and sends a summary through SNS.
- **Read replica:** a second copy of the ERP database that can be promoted to a standalone instance if the primary is lost (promotion breaks replication, so the replica becomes the new primary).
- **EFS automatic backups:** enabled for the shared documents.
- **Golden AMI:** an image of the ERP server from which a replacement can be launched.
- **Restore:** a snapshot is restored to a new instance and the data is verified with the MySQL CLI and `mysqldump`.

Test scenario

1. Write a distinctive note from the web page.
2. Run the Lambda and wait for the snapshot to become Available.
3. Delete the notes (simulated disaster).
4. Restore from the snapshot and confirm the note is back.

RPO is the time since the last snapshot (for point-in-time restore, RDS can recover to within minutes of the failure). **RTO** is the time needed to restore the instance and depends on database size.



Figure 9: RDS snapshots of erp-mysql-db on 2 October 2026: the manual snapshot (erp-db-manual-snapshot-v1) and the snapshot created by the Lambda function (auto-lambda-erp-mysql-db-...).

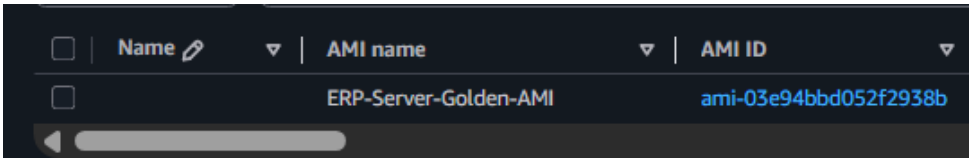


Figure 10: Golden AMI of the ERP server (ERP-Server-Golden-AMI).

9. High availability and scalability

Mechanism	Benefit
Application Load Balancer	Distributes traffic across healthy application instances
Auto Scaling Group	Maintains required capacity and scales between 2 and 4 instances
Multiple Availability Zones	Reduces the impact of an AZ failure
Amazon EFS	Shared storage for multiple application servers
RDS read replica	Second copy of the ERP data; offloads reads and can be promoted
Transit Gateway	Scalable, centralized connectivity between environments
CloudWatch and SNS	Alarm-based visibility and notifications

10. Cost optimization

- RDS on db.t3.micro: a Single-AZ primary and a single small read replica (the replica adds roughly the cost of a second db.t3.micro instance).
- Free S3 Gateway Endpoint.
- EFS lifecycle policy moves files to Infrequent Access after 30 days and to Archive after 90 days.
- Only the latest 3 manual snapshots are retained.

Transit Gateway cost estimate

Item	Rate (verify for ap-south-1)	Estimate
VPC attachments (3)	about $0.05 per attachment-hour	about $110 per month (3 x 730 h x $0.05)
Data processing	about $0.02 per GB sent to the TGW	negligible at the low inter-VPC traffic of this project

Rates are indicative; check the current ap-south-1 price in the AWS Pricing Calculator before budgeting.

Trade-offs. Single-AZ RDS means possible downtime if the AZ fails (covered by snapshots). The ASG servers run in public subnets, with access restricted by security groups; note that AWS also charges for public IPv4 addresses. The Transit Gateway is the largest fixed cost and was accepted for routing control and scalability (section 4.4).

11. Testing and expected results

Test	Expected result
Open the domain and log in	Welcome page loads; notes are saved in RDS
Ping or SSH from the Bastion to ERP and App servers	Works through the Transit Gateway
Try to connect from the ERP VPC to the App VPC	Blocked (isolation by TGW route tables)
Refresh the ALB DNS repeatedly	Server hostname changes (load balancing)
Write a file to EFS on one server	File is visible on another server
Generate CPU load with stress-ng	New instance launches and an alert e-mail arrives
Terminate a server manually	ASG replaces it automatically
Run the Lambda, then restore	Snapshot appears and data is recovered

12. Deliverables map

Deliverable	Section
Architecture description	3
Network, VPCs and Transit Gateway	4
IAM groups and security	5
ERP multi-tier system and internal application	6
Monitoring and alarms	7
Disaster recovery	8
High availability	9
Cost optimization	10
Testing	11

13. Production improvements

- RDS Multi-AZ, and application servers in private subnets behind NAT.
- HTTPS using ACM on the ALB.
- Passwords and credentials stored in AWS Secrets Manager.
- AWS Systems Manager Session Manager instead of a Bastion Host with open SSH.
- Scheduled backups through EventBridge, and cross-region copies for secondary-region disaster recovery.
- Infrastructure as Code with Terraform or CloudFormation.

14. Conclusion

This project delivers a segmented, scalable and highly available AWS environment for a startup. Separate VPCs and IAM groups isolate the Admin, ERP and App teams; the Transit Gateway connects them centrally while keeping ERP and App isolated from each other; and the load balancer, Auto Scaling Group and shared EFS storage keep the application tier resilient. CloudWatch, SNS, snapshots and the Golden AMI give the operations team visibility and a recovery path, with cost-conscious choices throughout.

All passwords and sensitive endpoints were intentionally omitted from this document.